

Class 10 NCERT Maths

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Chapter 9: Some Applications of Trigonometry (Core-Basics Edition)

CHAPTER 9: SOME APPLICATIONS OF TRIGONOMETRY

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) TRIG RATIOS (Chapter 8 recap)

$\sin = \text{Perp/Hyp}$, $\cos = \text{Base/Hyp}$, $\tan = \text{Perp/Base}$.

2) SPECIAL ANGLE VALUES (zaroori)

$\tan 30 = 1/\sqrt{3}$, $\tan 45 = 1$, $\tan 60 = \sqrt{3}$.

$\sin 30 = 1/2$, $\sin 45 = 1/\sqrt{2}$, $\sin 60 = \sqrt{3}/2$.

3) sqrt VALUES

$\sqrt{3} = 1.732$, $\sqrt{2} = 1.414$ (decimal answer ke liye).

4) HORIZONTAL AUR VERTICAL

Horizontal = leti (zameen ke parallel).

Vertical = khadi (tower/building ki height).

5) RIGHT TRIANGLE BANTA HAI

Tower (vertical) + zameen (horizontal) + line of sight
milke right-angled triangle banate hain.

6) ALTERNATE ANGLES (parallel lines)

Do horizontal lines parallel ho toh angle of
elevation = angle of depression.

7) SOLVE FOR UNKNOWN

$\tan(\text{angle}) = \text{opposite/adjacent}$. Ek pata ho toh
doosra nikaal lo (multiply/divide).

NOW: ASLI CHAPTER 9 SHURU

TOPIC 1: LINE OF SIGHT

Line of sight = aankh se object tak ki seedhi line.

TOPIC 2: ANGLE OF ELEVATION

Object UPAR ho (jaise building top), horizontal se upar wala angle = ANGLE OF ELEVATION (upar dekhna).

TOPIC 3: ANGLE OF DEPRESSION

Object NEECHE ho (jaise cliff se boat), horizontal se neeche wala angle = ANGLE OF DEPRESSION (neeche dekhna).

[CORE] Angle of elevation = angle of depression
(alternate angles, RED page point 6).

TOPIC 4: SOLVING APPROACH

EXAM ALERT: Har question ka same tareeka:

1. Diagram banao (right triangle)
2. tan/sin/cos choose karo (zyadatar tan)
3. Equation banao aur solve karo

$\tan = \text{height} / \text{base}$ (sabse zyada use hota).

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Do angles (2 positions)

(Exam favourite)

Tower ke base se kuch door se top ka elevation 30 hai.
20 m paas aane par 60 ho jaata. Tower ki height?

Let height = h, paas wali distance = x.

$$\tan 60 = h/x \rightarrow h = x \sqrt{3} \quad \dots(1)$$

$$\tan 30 = h/(x+20) \rightarrow h \sqrt{3} = x+20 \quad \dots(2)$$

$$(1) \text{ ko } (2) \text{ mein: } (x \sqrt{3}) \sqrt{3} = x + 20$$

$$3x = x + 20 \rightarrow x = 10$$

$$h = 10 \sqrt{3} = 17.32 \text{ m.}$$

Solved Example 2 - Depression (boat)

(Exam favourite)

20 m unchi cliff ke top se boat ka depression 30.

Boat kitni door?

$$(\text{depression} = \text{elevation} = 30)$$

$$\tan 30 = 20/d \rightarrow 1/\sqrt{3} = 20/d$$

$$d = 20 \sqrt{3} = 34.64 \text{ m.}$$

Solved Example 3 - Kite (string length)

(Exam favourite)

Patang ki dori 100 m, ground se 60 angle. Patang ki height?

$$\sin 60 = h/100 \rightarrow h = 100 \times (\sqrt{3}/2) = 50 \sqrt{3}$$

$$h = 86.6 \text{ m.}$$

Solved Example 4 - Simple tower (elevation 60)

Base se 30 m door, top ka elevation 60. Tower height?

$$\tan 60 = h/30 \rightarrow h = 30 \sqrt{3} = 51.96 \text{ m.}$$

Solved Example 5 - Find distance (elevation 45)

15 m unchi building ka top, ground point ka elevation
45. Point kitni door?
 $\tan 45 = 15/d \rightarrow 1 = 15/d \rightarrow d = 15 \text{ m.}$

Solved Example 6 - Ladder (cos)

Ladder ground se 60 angle, foot wall se 2.5 m door.
Ladder length?
 $\cos 60 = 2.5/L \rightarrow 1/2 = 2.5/L \rightarrow L = 5 \text{ m.}$

Solved Example 7 - Tower height (45, base 40)

Base se 40 m door, elevation 45. Height?
 $\tan 45 = h/40 \rightarrow 1 = h/40 \rightarrow h = 40 \text{ m.}$

Solved Example 8 (EASIEST) - Concept

Angle of elevation aur depression mein fark?
Elevation = object upar (upar dekhte).
Depression = object neeche (neeche dekhte).

Q AND A TIME

Q1. CORE: angle of elevation aur depression ek line mein.

Q2. Tower base se 50 m door, elevation 45. Height nikaal.

Q3. 30 m cliff top se boat depression 30. Boat door?
($\tan 30 = 1/\sqrt{3}$)

Q4. Patang ki dori 200 m, ground se 30 angle. Height?
($\sin 30 = 1/2$)

Q5. Tower 60 m. Base se 60 m door se top ka elevation?
($\tan = 60/60 = 1 \rightarrow \text{angle?}$)

Q6. Tower base se kuch door se elevation 30. 10 m paas
aane par 60. Height nikaal.

Q7. CORE CHECK: $\tan 60$ ki value kya hai?

SUMMARY

1. Line of sight: aankh se object tak seedhi line.
2. Elevation: object upar; Depression: object neeche.
3. Elevation = Depression (alternate angles).
4. Steps: diagram -> tan/sin/cos choose -> solve.
5. $\tan = \text{height/base}$ (zyadatar use hota).

CORE (RED page) revise: trig ratios, special angles, sqrt values, alternate angles, right triangle banna.